

Hardware CWE SIG Meeting Notes

Meeting Minutes

August 21, 2026

Meeting Attendance

Bob Heinemann	Ashrafi Gulam	Rachana Maitra
Steve Christey Coley	Mohammed	Jason Oberg
Andreas Schweiger	Mohan Lal	Irena Bojanova
Parbati K. Manna	Bruce Monroe	Jeremy Lee
William Ferguson	Amitabh Das	Paul Wortman
Mitchell Poplingher	Sohrab Aftabjahani	

Agenda

- RTL WG, HW View, and Other Updates
- HW Lifecycle Discussion
- Memory Weaknesses – deferred

Meeting Notes

RTL WG, HW View, and Other Updates

Housekeeping and Release Updates

Bob announced that the next Hardware CWE SIG meeting is September 18 and said CWE 5.0 is planned for the October time frame. He also shared an update on the CWE content repository.

Hardware View flattening and relationship updates

Bob reported that the Hardware View flattening discussed in prior meetings is in the development environment and planned for CWE 5.0. The flattening presents affected

entries at the same visible level while retaining their underlying relationships, which remain available in the entries and in the hierarchical research view. He also said previously discussed PeerOf and CanAlsoBe relationship updates are planned for the same release.

RTL working group update

Bob said the RTL working group had provided input across multiple sections of the simulation-synthesis mismatch entry. He is compiling and reconciling those updates for review at the working group meeting on August 24. He expects the entry to be included in CWE 5.0.

HW Lifecycle Discussion

Purpose and CWE lifecycle framing

Bob framed the discussion around whether the current CWE lifecycle phase vocabulary adequately represents hardware concerns and, if changes are useful, what level of additional precision would help. He emphasized that CWE phases are metadata tags rather than a process model: phase labels are independent and unordered, and are used to indicate where weaknesses may be introduced and where mitigations may be applied.

Current phase use and analysis caveats

Using current Hardware CWE content, Bob showed that Modes of Introduction and Potential Mitigations are concentrated largely in the Architecture and Design and Implementation phases. He said the supporting corpus analysis and charts were AI-assisted, preliminary, and not fully verified, although spot checks appeared representative. Steve said he reviewed the analysis code and believed the algorithm appeared accurate despite poor maintainability.

Bob also noted data-quality limitations in the corpus. More than 100 notes elements have phase values but no explanatory text, so some hardware themes may be missing from the analysis. He also cautioned that some testing-related material may reflect detection methods rather than mitigation and may warrant separate quality review.

Sohrab asked where the current lifecycle vocabulary came from and how users can determine what each phase means. Steve explained that the vocabulary evolved over CWE's history through community contributions and gap-filling rather than from a single formal source, and that the schema lists the phase values without fully defining each one. Sohrab suggested clearer documentation and noted that combined software-oriented terms do not always map cleanly to hardware practice. Steve clarified that the groupings

shown in Bob's slide were informal presentation groupings and are not represented in the CWE schema.

External lifecycle references

Daniel said the SAE G-32 Hardware Assurance Committee has developed lifecycle stages and definitions and offered to obtain that material for the SIG. Bob said those definitions could be helpful for aligning the work. Later, Sohrab suggested reviewing publicly available lifecycle models from major EDA companies and semiconductor vendors rather than defining hardware terms from scratch, and offered to collect material that can be shared.

Pre-silicon and post-silicon concepts

Jason suggested considering a split between pre-silicon and post-silicon activities. Rachana supported recognizing pre-silicon and post-silicon concepts but viewed them as cross-cutting characteristics rather than necessarily separate lifecycle phases. She noted that some activities, such as manufacturing, are inherently post-silicon, while verification and testing can occur before or after silicon.

Abraham questioned the usefulness of post-silicon as a standalone phase, especially for weakness introduction, and cautioned against adding coarse labels that could reduce data quality. Steve noted that phase values cover both introduction and mitigation, so a phase may still be relevant even if weaknesses are not commonly introduced there. Irena added that weaknesses can also emerge after manufacture, and that pre-silicon and post-silicon concepts may fit within existing hardware-relevant phases rather than replacing them.

Architecture, design, implementation, and integration

Arun described a hardware development flow in which requirements state security objectives; architecture selects higher-level mechanisms; design selects specific structures and implementations; implementation realizes those choices in RTL; and integration combines IP and supporting logic into larger hardware systems. He supported separating Architecture and Design because the two terms represent meaningfully different activities in hardware development. His description also placed RTL implementation and integration before manufacturing, not in a post-silicon implementation phase.

Verification, validation, testing, and release terminology

Rachana questioned the boundary between manufacturing and testing and asked whether a recognizable release concept might be useful. Steve noted that Distribution exists in the current vocabulary and agreed that terminology recognizable as release could make sense, but no change was adopted. Sohrab later distinguished validation from manufacturing test,

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explaining that validation checks whether the design matches its specification while manufacturing testing checks whether the produced device was manufactured correctly. He suggested that pre-silicon and post-silicon distinctions may be most useful for validation or verification rather than for every hardware phase.

Future AI-assisted hardware implementation

Manna raised the possibility that future RTL and other hardware design artifacts may increasingly be generated by AI, potentially changing the abstraction level at which implementation errors are introduced. Arun responded that an AI-generated weakness would still be introduced within an existing lifecycle activity such as implementation. Manna agreed with the immediate point while noting that the semantics of implementation may evolve as design abstraction changes.

Discussion outcome

Bob closed the discussion by asking whether the SIG should leave the current phase vocabulary unchanged or continue exploring hardware-relevant updates. No participant indicated that the work should simply be abandoned. A subsequent show-of-hands indicated support for continuing the discussion, while Manna cautioned against creating too many phases and Bob observed that some participants appeared indifferent. The meeting did not approve a specific set of new phase values.

Memory Weaknesses - deferred

Deferred discussion

Bob said the memory-weakness topic would be deferred because the hardware lifecycle discussion was expected to take the available meeting time. No substantive memory-weakness discussion occurred during this meeting.

Decisions Made

1. Continue the Hardware CWE lifecycle-phase discussion rather than treat the current phase vocabulary as final. No specific new phase set was approved at this meeting.
2. Defer the memory-weakness discussion from this meeting so the hardware lifecycle discussion could use the available time.

Action Items:

1. Reconcile RTL working group updates and prepare the simulation-synthesis mismatch entry for review at the August 24 working group meeting. (Bob)

2. Obtain and share SAE G-32 Hardware Assurance lifecycle stage definitions for SIG consideration. (Daniel)
3. Gather and share publicly available hardware lifecycle material from major EDA companies and Intel. (Sohrab)
4. Share relevant hardware lifecycle materials on the mailing list to support the continuing phase discussion. (HW CWE SIG members)

Next Steps

Continue the lifecycle-phase discussion asynchronously, using the G-32 Cyber Physical Systems Security Committee material, industry lifecycle references, and the meeting's hardware-specific distinctions to refine candidate terminology without turning CWE into a full development-process model.

Consider whether lifecycle phase terms need clearer definitions and whether testing, validation, verification, and detection content in Hardware CWE should be reviewed for scoping or classification quality.

The RTL working group will review reconciled entry updates on August 24. The next Hardware CWE SIG meeting is September 18. The memory-weakness topic remains deferred for future discussion.